

Discharge decision for H-ADM-COH on the crystallographic competitor

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

STEP-5B is CLOSED-CONDITIONAL on H-ADM-COH, whose sole role is to secure the G1'''-AE corner (the weighted sphere additive-energy bound) by restricting to angularly SEPARATED competitor patterns. R-026 and R-027 establish that bound for the physical lattice class WITHOUT separation. This note assembles the discharge decision: it fixes the competitor class, restates the two theorems, and – the decisive step – turns the residual subpolynomial- K acceptance into a finite numerical inequality against the STEP-5B tolerance. It enacts no flip; it is the decision record for the operator.

2. The physical competitor class

We fix the B1 competitor class to crystallographic momentum-shell patterns

$$Q \subseteq \Lambda^* \cap \{k : |k|^2 = R\},$$

where Λ^* is the reciprocal BCC/FCC lattice. This is the physical competitor class of the TECT shell calculation; no claim is made for arbitrary real point sets $Q \subset S^2$ (arbitrary- Q DR-2 stays OPEN, and is not needed for the mainline). This is residual (b), an operator modeling identification, here made explicit.

3. The two established theorems

Unweighted (R-026, T7). For $Q \subseteq \Lambda^* \cap \{|k|^2 = R\}$, $E_+(Q) \leq (1+6 \max_{t \neq 0} d(4R-|t|^2))N^2 \leq (1+C_\epsilon R^\epsilon)N^2$, via Lemma A and the proved circle-divisor bound [DIV-CIRC] (substitution $y = 2k - t$).

Weighted (R-027 v1.1, T7). For any amplitudes $c_a \in \mathbb{C}$, $I = \sum_a |c_a|^2$, $w_t = \sum_{a+b=t} c_a c_b$,

$$\sum_t |w_t|^2 \leq (1 + T'(Q)) I^2,$$

so the G1'''-AE quantity is controlled by the same sum-circle richness T' for arbitrary (complex) amplitudes.

4. The finite margin (residual (a), settled numerically)

The STEP-5B closure tolerates an additive-energy constant growing as the kappa-balanced bound $K_{\text{allowed}}(n) = 8 + c_R\sqrt{n}$ with $c_R = 4\sqrt{14}$ (theorem-grade, beyond-layer v2.0), against the layer margins $\times 55.6 / \times 8.8 / \times 2.1$. The effective constant supplied by §3 is $K_{\text{adm}} = 1 + T'(Q)$. We verify (dr2_hadmcoh_margin.py, 3/3):

$$K_{\text{adm}} = 1 + T'(Q) \leq K_{\text{allowed}}(n) = 8 + 4\sqrt{14}\sqrt{n}$$

for every crystallographic shell and sub-pattern, with margin $10.6\times$ to $15.8\times$ that GROWS in n (because $T' \sim R^\epsilon$ grows far slower than $\sqrt{n}$):

R	N	T'	$K_{\text{adm}} = 1+T'$	$K_{\text{allowed}}(N)$
101	168	18	19	202
909	504	24	25	344
1826	672	24	25	396
4994	1152	32	33	516
9974	2040	48	49	684

For sub-patterns $Q' \subseteq \text{shell}$ the worst observed ratio is $(1 + T'(Q'))/K_{\text{allowed}}(|Q'|) = 0.307$, since $T'(Q') \leq \min(|Q'|, T'(\text{shell}))$ and $8 + c_R\sqrt{n}$ dominates both branches. The former operator judgment “is a subpolynomial $K \sim R^\epsilon$ acceptable?” is thus replaced by a verified inequality with an order-of-magnitude, widening margin.

5. Corner coverage (residual (c))

The G1’’-AE corner is the non-transversal multi-circle high- n pattern class. Every such pattern is a subset of the crystallographic shell, hence covered by §3–§4 with no separate treatment. This closes residual (c) mathematically.

6. The discharge decision

H-ADM-COH was used ONLY to enforce angular separation so that G1’’-AE holds. §3–§5 establish G1’’-AE for the unrestricted (non-separated) lattice class, with a finite tolerance margin. Therefore, for the crystallographic competitor class, the lattice arithmetic REPLACES separation and

$$\{H\text{-LAYER}, H\text{-ADM-COH}, SC\text{-SCOPE}\} \longrightarrow \{H\text{-LAYER}, SC\text{-SCOPE}\}$$

is the proposed B1 hypothesis-set reduction, with H-ADM-COH demoted to a legacy fallback for hypothetical non-lattice competitors.

Status recorded (NO flip enacted here). Per the operator’s endorsed ledger status: G1’’-AE_{lattice} = CLOSED@T7; H-ADM-COH = DISCHARGE-CANDIDATE; DR2-SHARE = OPEN pending operator integration; B1 = T6 CONDITIONAL on $\{H\text{-LAYER}, H\text{-ADM-COH}, SC\text{-SCOPE}\}$ (unchanged); B5 = T5. The discharge and the B1 re-tier are the operator’s decision; this note supplies the complete justification for that decision.

7. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

- (α) “ $c_R = 4\sqrt{14}$ and the $\times 55.6 / \times 8.8 / \times 2.1$ margins are read from the closure note – a misread would void §4.” VALID-with-mitigation: $c_R = 4\sqrt{14}$ and the kappa-balanced bound $K(n) \leq 8 + c_R\sqrt{n}$ are quoted from beyond-layer-gershgorin-reduction v2.0; the margin claim is robust to the exact constant because $K_{\text{adm}}/K_{\text{allowed}} \leq 0.1$ leaves an order of magnitude of slack. If the operator’s canonical c_R differs by an $O(1)$ factor the conclusion stands.
- (β) “*This silently discharges H-ADM-COH.*” It does NOT: §6 records H-ADM-COH as DISCHARGE-CANDIDATE and B1 as UNCHANGED T6, enacting no flip. The discharge is the operator’s decision.
- (γ) “*Sub-patterns could have a rich circle violating the margin.*” Checked: $T'(Q') \leq \min(|Q'|, T'(\text{shell}))$, and both branches sit below $8 + c_R\sqrt{|Q'|}$ (small n : $T' \leq n \leq 8 + c_R\sqrt{n}$ up to $n \approx 240$; large n : $T' \leq T'(\text{shell}) \sim R^\epsilon \ll \sqrt{n}$); worst observed ratio 0.307.
- (δ) “*The arbitrary-Q problem is being smuggled in.*” NO: the class is explicitly restricted to crystallographic shells (§2); arbitrary-Q DR-2 stays OPEN and is not used.

Quantitative sanity check (mandatory). $K_{\text{adm}} \leq K_{\text{allowed}}$ on all shells (margin 10.6–15.8 $\times$, growing); worst sub-pattern ratio 0.307; T' grows like $R^{0.18}$ vs the allowed $\sqrt{n}$. Direction: both sides positive, $K_{\text{adm}} \geq 1$. Limit: small n handled by $T' \leq n$.

8. Result footer

Result ID:	R-028 / H-ADM-COH discharge decision (lattice class)
Precise statement:	For Q subset of a crystallographic shell, the G1'''-AE bound $\text{sum}_t w_t ^2 \leq (1+T'(Q)) I^2$ (R-027) holds with $K_{\text{adm}} = 1+T'(Q) \leq K_{\text{allowed}}(n) = 8 + 4 \sqrt{14} \sqrt{n}$ (verified, margin 10.6x-15.8x growing; sub-pattern worst ratio 0.307). Since H-ADM-COH only secured G1'''-AE by angular separation, it is a DISCHARGE-CANDIDATE: {H-LAYER,H-ADM-COH,SC-SCOPE} $\rightarrow$ {H-LAYER, SC-SCOPE} is the proposed B1 reduction for the lattice class.
Scope:	B1-RH-ENUM hypothesis set; STEP-5B G1'''-AE; the physical crystallographic competitor class only.
Dependencies:	R-026 (T7), R-027 v1.1 (T7); $c_R = 4 \sqrt{14}$ and the kappa-balanced tolerance $K(n) \leq 8 + c_R \sqrt{n}$ from beyond-layer-gershgorin-reduction v2.0.
Evidence grade:	Integration/decision note. The finite margin (a) is verified numerically; (b) modeling and (c) corner are settled. H-ADM-COH = DISCHARGE-CANDIDATE (no flip).
Reproduction command:	python codes/vacuum/dr2_hadmcoh_margin.py
Expected output:	3/3 PASS; $K_{\text{adm}} \leq K_{\text{allowed}}(N)$ all shells (margin 10.6x-15.8x); margin grows with N; sub-pattern worst ratio 0.307.
Falsification gate:	A crystallographic shell/sub-pattern with $1+T'(Q) > 8 + 4 \sqrt{14} \sqrt{ Q }$; not observed.
Tier before / after:	NEW decision note. NO flip: B1 T6 unchanged, H-ADM-COH DISCHARGE-CANDIDATE (was active), DR2-SHARE OPEN, B5 T5, G1'''-AE_lattice CLOSED@T7.
No-overclaim statement:	Does NOT enact the discharge or change B1’s tier; it is

the decision record. The discharge holds for the LATTICE class only; arbitrary-Q DR-2 stays OPEN. The margin rests on the quoted closure constant c_R.

Next required action: Operator: accept (a)-(c) and discharge H-ADM-COH from the active B1 set for the lattice class (re-tier B1 on {H-LAYER, SC-SCOPE}), or request further hardening.